

AI-Based Urban Ecosystem Health Intelligence System (UEHIS)



1. The Unified Concept (How it all connects)

This project is an end-to-end intelligence system that transforms raw satellite data into actionable conservation metrics and carbon offset data for the SEER Bharat Foundation. It operates in four sequential phases:

- **Phase 1: Urban Landscape Mapping (The "What is there?")** The system ingests multi-spectral satellite data (Sentinel-2, Landsat) and uses semantic segmentation to map the exact locations of existing trees and surrounding infrastructure (concrete vs. open soil).
- **Phase 2: 3D Canopy Simulation (The "How does it interact?")** The API extrapolates this 2D map into a 3D spatial environment. It models the existing canopy volume, building heights, and calculates the exact shadow geometry and thermal interactions in that specific microclimate.
- **Phase 3: The 8-Factor Urban Ecosystem Health Index (UEHI)** Using data from Phase 1 & 2 combined with external APIs, the machine learning model generates a composite 0-100 Health Score for **every single city block**. The AI calculates this using an 8-parameter feature matrix:
 - **Vegetation & Ecology:** *NDVI* (Canopy health), *Tree Density* (Trees per sq meter), and *Biodiversity* (Species variance & habitat connectivity).
 - **Climate & Air:** *Local Temperature* (Urban Heat Islands) and *AQI* (Particulate matter levels).
 - **Infrastructure & Survival:** *Impervious Surfaces* (Concrete suffocating roots) and *Water Availability* (Groundwater recharge potential).
 - **The Human Factor:** *Population Exposure* (How many vulnerable people actually live in that specific block).
- **Phase 4: Carbon Sink Planner (The "How do we optimize & fund it?")** Using the 3D volume (biomass) calculated in Phase 2 and the health score from Phase 3, the AI calculates the exact tonnage of CO2 sequestered by the existing forest. Furthermore, it acts as a "Planner" by running optimization algorithms to recommend exactly where *new* trees should be planted to maximize future carbon absorption and bridge ecological gaps.

2. The Societal Impact (Why the NGO needs this)

This unified system shifts the NGO from **reactive planting** to **proactive ecosystem and financial management**.

- **Targeted Interventions:** If a city block scores a 40/100 on the UEHI because of high *Population Exposure* and high *Impervious Surfaces*, the NGO knows exactly what to do. They must prioritize this block to protect the high population, and they must de-concretize the roots of the existing trees to save them.
- **Unlocking CSR Funding:** The Carbon Sink Planner is a financial game-changer. The NGO can take this automated report to corporate donors and say, "*Fund our drive in this specific ward*,"

and our AI proves it will offset your company's carbon footprint by 45 tons of CO2 this year." It provides data-driven proof for funding.

3. The Technical Architecture (Why the Professor will love it)

This is a highly rigorous, enterprise-grade system design:

- **Data Ingestion & Feature Engineering:** Automated fetching of 8 distinct open-source geospatial datasets (Earth Engine, OpenStreetMap, AQI APIs, Census grids).
- **Machine Learning:** Combining Computer Vision (Semantic Segmentation for mapping) with a multi-variate regression/classification model to generate the block-by-block UEHI score.
- **Cloud Infrastructure:** The entire 4-stage engine is decoupled into a REST API microservice, containerized via Docker, and deployed to the cloud, allowing the NGO's web dashboard to query it seamlessly.